How secure are cloud-based systems?

Cyber-attack affects hundreds of Bealand businesses

Cloud computing is the delivery of computing services, including servers, storage, databases, networking, software, analytics and intelligence over the internet ('the cloud'). The latest international cyber-attack, which occurred last week, was aided by the access to such cloud computing systems. The attackers were able to access millions of cloud accounts and corrupt and steal files and personal information from more than 100 large companies and government departments throughout Bealand.

This was not the first cloud services cyber-attack, and it certainly will not be the last. Cloud system weaknesses were also to blame for last year's breach at Cazpa, Bealand's second largest bank. In this case, a cloud system weakness led to the corruption of millions of customer records, including their bank account information.

These attacks on cloud service providers by criminal hackers are increasing as cloud computing has become more popular worldwide as the predominant model for information systems management. Cloud is increasingly the default mode of computing for many Bealand businesses, meaning ever more users and critical data rely on these systems. Therefore, cloud computing is an important source of risk because it has quickly replaced traditional IT storage systems and importantly, because it has concentrated ownership of cloud systems design, management and control within a very small number of cloud service provider businesses in Bealand.

One might assume that competition between cloud service provider businesses would incentivise them to invest in security measures. However, security is not a priority for these businesses because the losses from data breaches are largely borne by their customers. As long as a cloud service provider is not losing significant customer numbers (which research indicates does not usually happen after a security incident), there is seemingly limited incentive to invest significantly in additional security. In addition, research indicates that the cloud service providers do not face any consequences from investors as share price declines after a cloud security breach are often both small and temporary.